2024 8th International Conference on Computational System and Information Technology for Sustainable Solutions (CSITSS) | 979-8-3315-0546-2/24/$31.00 ©2024 IEEE | DOI: 10.1109/CSITSS64042.2024.10816946

State-of-the-Art Deep Learning Strategies for Multi-Label Classification of Chest X-Ray Thoracic Disorders

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***Abstract*—This study tackles the challenge of multi-label classification of chest X-rays from the NIH dataset, focusing on diagnosing 14 thoracic pathologies. It employs advanced deep learning models and ensemble techniques to enhance diagnostic accuracy. Three key CNN architectures are used: MobileNet, EfficientNet, and Extreme Learning Machine (ELM). MobileNet is efficient for resource-limited environments, EfficientNet balances accuracy and computational efficiency, while ELM offers fast learning for large datasets. Ensemble methods like average, weighted average, and max vote combine these models to boost robustness and reliability. The performance is evaluated using metrics such as AUC, F1 score, and Hamming loss, with EfficientNet emerging as the top performer. The weighted average ensemble achieves the best results with an AUC of 0.8721, an F1 score of 0.8245, and a Hamming loss of 0.0987. This research demonstrates the potential of CNNs and ensemble methods in medical image analysis, advancing diagnostic tools for better clinical outcomes and improved healthcare.**

***Index Terms*—multi-label classification, MobileNet, EfficienNet, ELM, Ensemble learning, F1 score, Hamming loss.**

I. Introduction

The thorax, or chest, is the upper trunk between the neck and abdomen, protected by the rib cage, spine, and shoulder girdle. This area houses vital organs like the lungs, heart, and esophagus, essential for breathing and cardiovascular functions. Chest pain often leads to emergency visits, [1]requiring accurate diagnostics. Chest X-rays (CXRs) are common for diagnosing thoracic diseases, but their interpretation demands high expertise and involves uncertainty.To improve diagnostic accuracy and efficiency, Computer-Aided Diagnosis (CAD) systems offer a "second opinion" by providing preliminary analysis and identifying potential abnormalities. Advances in Artificial Intelligence (AI) and Computer Vision (CV) have enhanced CAD systems, particularly in medical image classification, which assigns diagnostic labels to images to detect diseases.

Historically, image classification [1] involved a two-stage process: feature extraction and classification. Deep learning, particularly Convolutional Neural Networks (CNNs), revolutionized this field by automating feature extraction and classification from raw images. CNNs, inspired by the human visual system, excel in capturing complex patterns.Early CNN models like LeNet, introduced in the 1990s, were successful in digit recognition. However, they require large labeled datasets, posing challenges in domains with limited annotated data. [2]Transfer learning addresses this by fine-tuning pre-trained models for specific tasks.Multilabel classification, assigning multiple labels to each instance, is crucial in medical imaging for detecting co-existing conditions from a single CXR. This can be achieved by transforming the multilabel problem or adapting classifiers to handle multilabel data directly.

In our study, we introduce an innovative method for identifying thoracic diseases from chest X-rays using transfer learning and multilabel classification. [3] We utilize classifiers such as MobileNet, DenseNet, and ELM to extract meaningful features from CXRs, followed by applying multilabel classification techniques to assign diagnostic labels. Our approach is validated on the widely used ChestX-ray14 dataset, demonstrating superior performance compared to [4] existing methods. Our results show state-of-the-art performance, with a significantly higher average AUC than previously reported, indicating its potential to enhance diagnostic accuracy in clinical practice.

The following section explores previous studies con-

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cerning the use of machine learning in multilabel classification of ChestX-ray14 images.

II. BACKGROUND STUDY

In this section, we discuss the related works about the Classification of OCT Images using Deep Learning Technologies. The paper [5]introduces an innovative approach for multi-label classification of common thorax diseases using chest X-rays. It combines CNNs for image feature extraction with problem transformation techniques for multi-label classification. Tested on ChestX-ray14 and CheXpert datasets, the method surpasses existing state-of-the-art approaches. It highlights the benefits of transfer learning in enhancing CAD system performance in medical image analysis, showcasing the effectiveness of deep learning for improving diagnostic accuracy. This paper [6], [7]introduces a hierarchical multi-label classification (HMLC) method for chest X-ray (CXR) CAD, emphasizing the integration of clinical taxonomies to enhance classification accuracy and generate meaningful predictions. Evaluated on the PLCO dataset, the HMLC approach achieves a mean AUC of 0.887, setting a new benchmark. It effectively addresses the challenge of incomplete labeling in medical datasets, contributing significantly to the advancement of CXR CAD by improving diagnostic performance and managing missing data. The paper [8]introduces a two-stream collaborative network (TSCN) for multi-label chest X-ray classification, utilizing lung segmentation. TSCN features a global branch for extracting image features and a local branch for lung-specific features. A self-adaptive weighted fusion scheme, based on the Hilbert-Schmidt independence criterion (HSIC), dynamically balances these features. Tested on the ChestX-ray14 dataset, TSCN achieves superior performance over previous methods. Ablation studies confirm the effectiveness of its components, and the method is noted for its efficiency in handling multi-label classification tasks. Overall, it presents a novel approach for improving CXR image classification. Recent studies [9]in automated chest disease diagnosis using deep learning have explored diverse methods to enhance accuracy and efficiency. Ganesan et al. used a Keras framework with depth-wise separable convolutions to classify lung diseases with 86.14% accuracy. Rajpurkar et al. applied unsupervised segmentation to annotate DRRs for multi-organ X-ray analysis. Salehinejad et al. developed a binary classifier for pneumonia detection from frontal chest X-rays. Irvin et al. employed DCGANs to generate synthetic chest X-rays, augmenting real data for improved accuracy. Chandra et al. combined multiple models achieving 95.63% accuracy in pneumonia detection. These studies highlight neural networks' role and emphasize the need for large, balanced datasets in improving cardiothoracic disease diagnosis. This paper [10], [11] reviews challenges and advancements in multi-label classification of chest X-ray images, focusing on issues such as visual similarity among lung pathologies and dataset imbalance. Previous studies used transfer learning with ResNet and DenseNet, achieving good results in single or dual disease detection but struggling with multi-disease scenarios. The paper proposes a new error function, Multi-label Softmax Loss (MSML), to handle multiple labels and imbalance effectively. Evaluated on ChestX-ray14, the MSML-based model shows improved AUC-ROC scores over existing methods, promising better performance in broader medical data classification tasks. This paper [12] comprehensively surveys deep learning applications in chest X-ray (CXR) analysis, covering 296 papers published before March 2021. It categorizes research by tasks such as image-level prediction, segmentation, localization, image generation, and domain adaptation. The authors discuss the impact of large, labeled CXR datasets on research trends, provide detailed insights into available datasets, and highlight the current state of the art. They also address challenges with public datasets, criteria for clinically useful systems, gaps in current literature, and explore commercial software for CXR analysis, along with future research directions. The paper [13] "Deep Transfer Learning for Multilabel Classification of Chest X-ray Images" explores using deep convolutional neural networks (CNNs) for multilabel classification of chest X-ray images. It leverages a private dataset of 1630 CXR images with disease labels and applies transfer learning with large open-source datasets to improve classification. The study evaluates different methods of reusing pretrained weights, varying sizes and similarities of source datasets, and different backbone CNN models. Results indicate that fine-tuning pretrained models generally yields superior predictions, and using multiple source datasets enhances performance compared to a single dataset. Overall, the research demonstrates that transfer learning can effectively boost prediction accuracy and reduce computational costs in CXR image analysis. The paper [14] "Deep Hierarchical Multi-label Classification of Chest X-ray Images" introduces a new method for detecting abnormalities in chest X-rays using hierarchical multi-label classification (HMLC). This approach is argued to be more suitable for medical imaging CAD than traditional flat multi-label methods, offering more meaningful and interpretable predictions aligned with clinical taxonomies. The model is tested on the PLCO dataset and achieves state-of-the-art performance with an average AUC of 0.887. Additionally, the authors propose a stable cross-entropy loss function for unconditional probabilities, further enhancing model accuracy. Overall, the paper underscores the potential of HMLC in medical imaging CAD, marking a significant advancement for chest X-ray analysis. This paper [15] reviews challenges and advancements in multi-label classification of chest X-ray images, focusing on issues such as visual similarity among lung pathologies and dataset imbalance. Prior studies used transfer learning with ResNet and DenseNet, achieving notable success in single or dual disease detection but facing challenges with multi-disease scenarios. The paper introduces a novel error

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function, Multi-label Softmax Loss (MSML), designed to effectively handle multiple labels and imbalanced data. Evaluated on the ChestX-ray14 dataset, the deep learning model incorporating MSML shows improved AUC-ROC scores compared to existing methods, suggesting promising avenues for enhancing performance in broader medical data classification tasks.

III. Methodology

In this part, we examine the classifiers employed in these studies, the proposed system, dataset characteristics, and ensemble techniques like average and weighted average ensemble are reviewed.

The method primarily comprises two main components: training classification models (Mobile-Net, DenseNet, ELM), and applying ensemble techniques across all three models. Figure 1 shows an illustration of the proposed system.

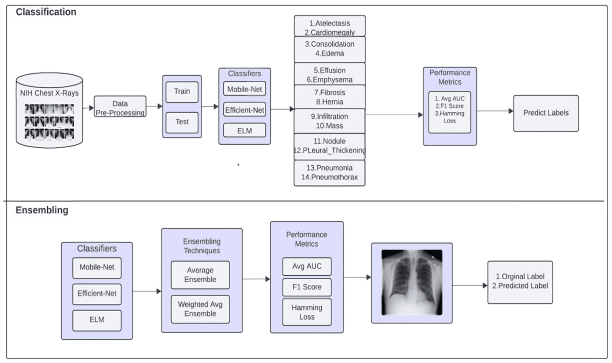


Figure 1. Proposed methodology of the system

A. Dataset Details

The dataset consists of 112,120 PNG-formatted, 1024x1024 frontal CXRs from 30,805 distinct patients. The fourteen main thorax diseases—Athelectasis, \cite{b14}Consolidation, Infiltration, Pneumothorax, Edema, Emphysema, Fibrosis, Effusion, Pneumonia, Pleural thickening, Cardiomegaly, Nodule, Mass, and Hernia—are identified with one or more on each CXR. CXRs with no illnesses found are reported as "No finding." Understanding dataset properties like size, format, and data dispersion requires visual exploration. The chest x-ray images with its different classes is shown in Figure 2.

Number of CXR, in CheXpert dataset, per pathology label,Hernia-227, Pleural thickening-3385,Fibrosis-1686,Emphysema-2516,Edema-2303,Consolidation-4667,Pneumothorax-5302, [?], [15]Pneumonia-1431,Nodule-6331,Mass-5782,Infiltration-19894,Effusion-13317,Cardiomegaly-2776,Atelectasis-11559 as shown in Figure 3.

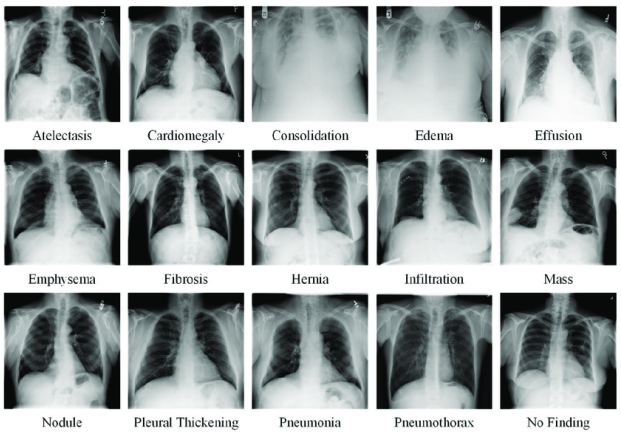


Figure 2. chest x-ray images with its classes

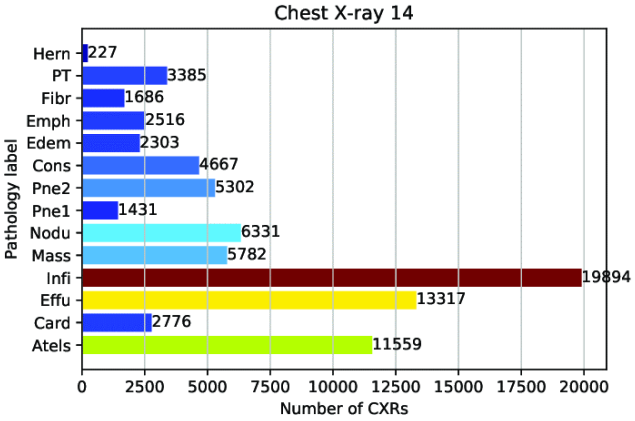


Figure 3. Number of CXRs, in ChestX-ray14 dataset, per pathology label

B. Data Pre-Processing

In our project using the ChestX-ray14 dataset for multilabel classification, thorough data preprocessing is crucial for optimal model performance. The dataset contains frontal CXRs from a diverse patient pool, necessitating standardization for consistency. Initially, we resize all images to a standardized 1024x1024 pixel format for uniform processing. Normalization techniques adjust pixel values to ensure consistency and minimize the impact of brightness or contrast variations. Additionally, augmentation methods, such as rotation, flipping, and zooming, enhance dataset diversity. These preprocessing steps mitigate biases and enhance our model's robustness to input data variations.

C. Feature Extraction

Feature extraction is a crucial stage in our multilabel classification methodology, focused on capturing pertinent information from chest X-ray images. Employing classifiers like MobileNet, DenseNet, and ELM, we derive significant features that encompass key attributes indicating various thoracic diseases. MobileNet and DenseNet are renowned for their capacity to represent features hierar-

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chically, while ELM provides efficient feature mapping abilities. Utilizing these classifiers, our goal is to extract discriminative features that accurately distinguish between different thoracic conditions in CXRs. These extracted features are then used as inputs for our multilabel classification model, enabling precise disease detection and classification.

D. Classifiers

In the context of multi-label classification, deep learning models like MobileNet, ELM, and DenseNet-121 typically utilize the sigmoid activation function in their output layers. Described by Formula 1, The sigmoid function, denoted by $\sigma(x)$ is a mathematical function commonly used in machine learning and neural networks. It takes an input x and maps it to a value between 0 and 1 ,e is the base of the natural logarithm. The sigmoid function is often used to model binary classification problems, where it can interpret input values as probabilities, with values closer to 1 indicating a positive outcome and values closer to 0 indicating a negative outcome. The function is defined as:

$$\sigma(x) = \frac{1}{1 + e^{-x}} \tag{1}$$

1) MobileNet: Depthwise separable convolutions are used in MobileNet, an effective convolutional neural network optimized for mobile and edge devices, to save computational costs. It can manage the complexity of predicting many labels per instance for multi-label classification by substituting a sigmoid layer for the final softmax layer, which allows it to estimate probabilities for each label independently. Because of this, MobileNet is a great option for environments with limited resources because it strikes a compromise between accuracy and efficiency while handling tasks that need several simultaneous classifications. Figure an overview of the architecture. .

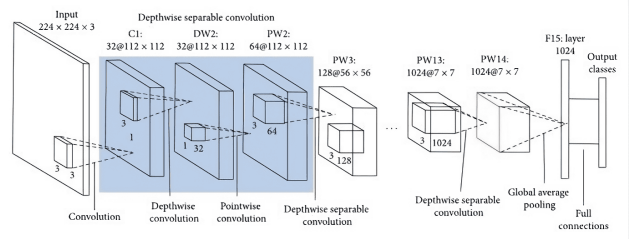


Figure 4. MobileNet Architecture

2) ELM: The Extreme Learning Machine (ELM) is a quick and efficient single-layer feedforward neural network that assigns weights and biases to hidden nodes at random, with just the output weights learned. Because of its quick training period and strong generalization capabilities, ELM can be used in real-time applications. By simultaneously learning to map input features to a set of labels, ELM can efficiently handle numerous outputs in multi-label classification, offering a rapid and simple solution for jobs requiring many concurrent predictions. See Figure 5 for an architectural overview.

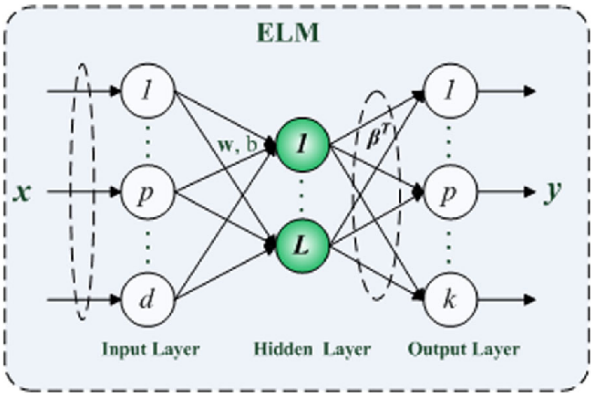


Figure 5. ELM Architecture

3) EfficientNet: The Google-designed EfficientNet family of convolutional neural networks uses a compound scaling technique that consistently scales depth, breadth, and resolution to produce great accuracy with little computing overhead. Because it can effectively handle the complexity of predicting many labels for each input, EfficientNet is very effective for a variety of tasks, including multi-label classification. Because of its scalable architecture, which strikes a compromise between efficiency and speed, it is perfect for deployment in settings where high precision is needed but resources are scarce.

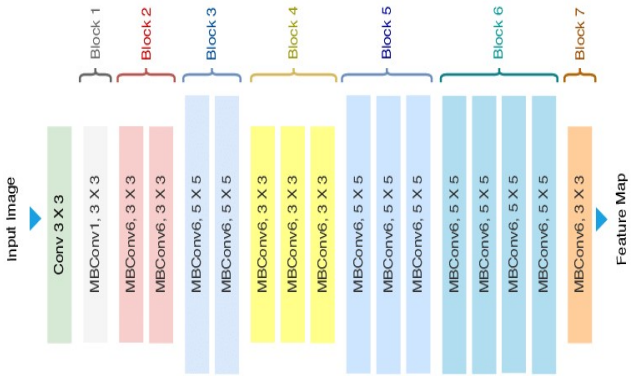


Figure 6. Efficient-Net Architecture

E. Ensembling

Ensembling with MobileNet, elm, and EfficientNet amalgamates diverse predictions utilizing average and weighted average methods, with the latter assigning weights based on individual model strengths. This nuanced approach aims to improve overall predictive performance.

1) Average Ensemble: The average ensemble merges results from several models to create a joint prediction determined by the highest sum. The ensemble prediction,

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denoted as $\hat{y}_{\text{avg_ensemble}}$, is derived by finding the argument that maximizes the sum of individual predictions $\hat{y}_i$, where i ranges from 1 to N, representing the total number of individual predictions. This approach, encapsulated by Formula 2, improves accuracy by combining diverse model outputs, consolidating their contributions into a unified decision.

$$\hat{y}\text{avg_ensemble} = \arg\max\left(\sum i = 1^N \hat{y}_i\right) \quad (2)$$

2) Weighted Average Ensemble: The ensemble with weighted averages integrates predictions with specified weights: 0.6 for CNN and 0.2 each for VGG-19 and DenseNet-121.The computation, delineated by Formula 3, derives the prediction $\hat{y}_{\text{weighted_avg_ensemble}}$ by summing the products of individual model predictions $\hat{y}_i$ and their corresponding weights w_i, where i ranges from 1 to N, The ensemble using weighted averages, assigning a greater weight to CNN, integrates perspectives from various models, representing the overall predictions.

$$\hat{y}\text{weighted_avg_ensemble} = \arg\max\left(\sum i = 1^N w_i \cdot \hat{y}_i\right) \quad (3)$$

3) Max Vote Ensemble: The max vote ensemble aggregates predictions by selecting the class with the highest frequency across the models. Each model casts a "vote" for a class, and the final prediction $\hat{y}_{\text{max_vote_ensemble}}$ is determined by choosing the class with the most votes, as described by Formula 4. For instance, if three models—CNN, VGG-19, and DenseNet-121—each predict a class for a given input, the class that receives the majority of votes becomes the final prediction. This method ensures that the ensemble reflects the consensus of the individual models, enhancing robustness and reducing the impact of individual model errors.

$$\hat{y}\text{max_vote_ensemble} = \arg\max\left(\sum i = 1^N \mathbb{1}(\hat{y}_i = c)\right) \quad (4)$$

IV. Results and Discussions

In this section, we examine the distribution of classes evaluate the effectiveness of deep learning models and assess the application of ensemble techniques to enhance prediction outcomes.

A. Individual Model Performance Analysis

1) MobileNet: MobileNet, trained on the NIH Lung dataset, demonstrated its performance through an ROC curve and a Confusion Matrix. The ROC curve showcases the trade-off between sensitivity and specificity, while the Confusion Matrix provides detailed insights into the model's predictions for various lung conditions. Evaluating these results helps assess MobileNet's accuracy and potential areas for improvement.

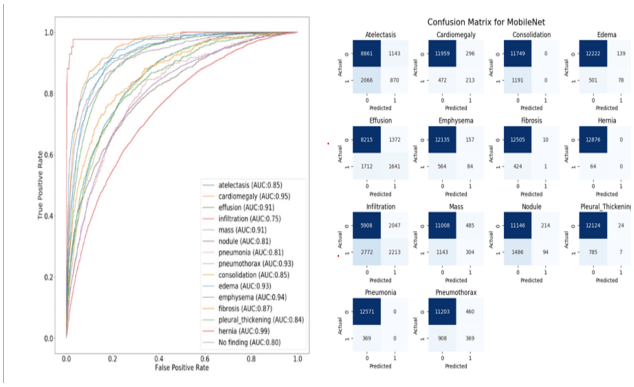


Figure 7. MobileNet ROC curve and Confusion Matrix

2) EfficientNet: This shows a multi-colored ROC curve on the left, illustrating sensitivity versus 1-specificity for different lung conditions like Atelectasis and Cardiomegaly. Each curve's AUC quantifies how well the model discriminates positive and negative cases, with higher values indicating better performance (e.g., Atelectasis with AUC 0.74 suggests moderate discrimination). On the right, a confusion matrix shows predicted versus actual conditions. True positives, true negatives, false positives, and false negatives are shown in rows and columns along with instances where the model correctly or incorrectly identified lung diseases. These visuals assess the EfficientNet model's ability to diagnose lung-related issues effectively.

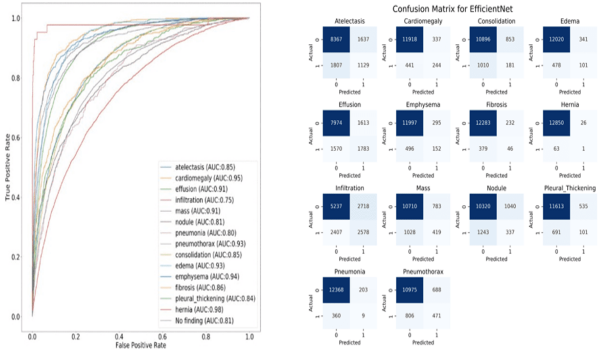


Figure 8. EfficientNet ROC curve and Confusion Matrix

3) ELM: This shows multiple ROC curves on the left, each representing different lung conditions like Asbestosis and Cardiomegaly. The AUC value next to each curve indicates the Extreme Learning Machine's (ELM) performance for that condition, with higher values indicating better discrimination. Sensitivity, Specificity, PPV (Positive Predictive Value), NPV (Negative Predictive Value), and Accuracy percentages for each condition are listed in the "Characteristics" table.

The "Confusion Matrix for ELM" displays predicted versus actual conditions, showing correct and incorrect predictions for each lung condition. This image offers

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insights into how well the ELM diagnoses lung-related issues.

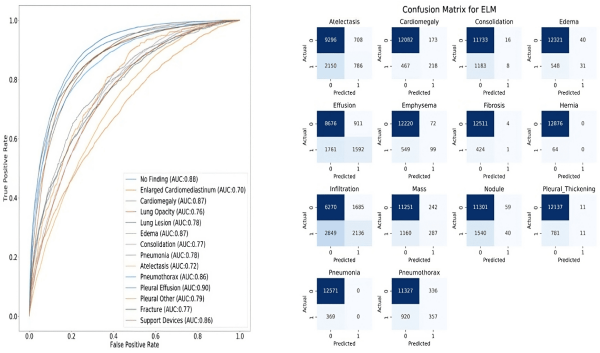


Figure 9. ELM ROC curve and Confusion Matrix

B. Performance Anaylysis

Three metrics—F1 score, Hamming loss, and Average AUC—are used in the table to compare the performance of several image categorization algorithms. The accuracy of a model on a classification job is shown by the F1 score. It takes into account the model's recall and precision. A metric for the disparity between two sets of data is called the Hamming loss. It can be used to quantify the difference between the ground truth labels and the expected labels in the context of picture classification. The average AUC, or area under the curve, is a metric used to assess how well a model can classify positive and negative examples.

Based on the table, the Weighted Average Ensemble model is performing the best on all three metrics. It has the highest F1 score (0.8245), the lowest Hamming loss (0.0987), and the highest Average AUC (0.8721). This suggests that the Weighted Average Ensemble model is more accurate at classifying images than the other models in the table

TABLE I
PERFORMANCE METRICS OF DIFFERENT MODELS

Model	F1 Score	Hamming Loss	Avg AUC
MobileNet	0.7105	0.1148	0.8054%
EfficientNet	0.8079	0.1056	0.8562%
ELM	0.6981	0.1571	0.7625%
Avg Ensemble	0.8112	0.1044	0.8419%
Wtd Avg Ensemble	0.8245	0.0987	0.8721%
Max Vote Ensemble	0.8094	0.1079	0.8271%

To summarize, the Weighted Average Ensemble outperforms other ensemble methods by strategically integrating the strengths of individual models while adapting to the unique characteristics of the dataset, ultimately resulting in enhanced predictive performance.

V. CONCLUSION AND FUTURE SCOPE

This study introduced a multi-label classification framework using transfer learning and ensemble techniques for detecting thoracic diseases from chest X-rays. Utilizing pre-trained models like MobileNet, DenseNet, and ELM, the approach outperformed existing methods on the ChestX-ray14 dataset, with the Weighted Average Ensemble showing the best results for practical use.

Future work could explore larger, more diverse datasets, integrate clinical data with X-rays to improve prediction accuracy, develop explainable AI to clarify model decisions, and validate the method in real-world clinical settings. These steps would enhance automated classification for better patient care.

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